

Why Stochastic is Sometimes Not Enough

Think about two very different businesses.

A biscuit company is fine with occasional stockouts. A customer who does not find their favourite biscuit will just buy another brand. For them, minimising average cost makes perfect sense.

Now think about a hospital stocking blood units before a major surgery. A stockout is not an inconvenience. It is life-threatening. The hospital does not care about the average outcome. They need to guarantee that even the worst case is survivable.

Robust optimization is built for the second kind of problem. It finds the plan that remains safe even when demand lands at its worst possible value.

There is another reason to use robust optimization. The stochastic model required you to know the probability of each scenario. But where do those probabilities come from? If your business is new, or demand is shifting, you may not have reliable probability estimates at all. Robust optimization only requires you to know the range demand can take. No probabilities needed.

The Uncertainty Set

In the stochastic model uncertainty was described by scenarios — a discrete list of possible demand futures. In robust optimization uncertainty is described by an uncertainty set — a continuous range that demand can take.

The simplest version is a box uncertainty set. For each period t , demand can be any value between a lower and upper bound:

$$d_t \in [d_{\text{bar}_t} - d_{\text{hat}_t}, d_{\text{bar}_t} + d_{\text{hat}_t}]$$

d_{bar_t} is the nominal demand. d_{hat_t} is the maximum deviation either side.

For example if average demand is 30 and the maximum deviation is 8, then demand could be anywhere from 22 to 38 in that period. You do not know where it will land. You just know it will not go outside that range.

The Gamma Parameter

If you protect against every period simultaneously being at worst case, you end up with an extremely expensive plan. In practice it is very unlikely that all periods hit worst case at the same time.

Gamma controls how many periods you assume can simultaneously be at their worst case. It is a number between 0 and T.

Gamma	What it assumes	Attitude
0	All periods at nominal demand	Optimistic — same as deterministic
1	At most 1 period at worst case	Mildly cautious
3	At most 3 periods at worst case	Balanced
5	At most 5 periods at worst case	Conservative
T	All periods at worst case simultaneously	Maximum safety — most expensive

Gamma is a dial. Turn it up for more safety. Turn it down to save money. Setting Gamma = 0 gives you the deterministic solution. Setting Gamma = T gives you the fully conservative solution that protects against everything.

The Price of Robustness Curve

When you solve the robust LP at every value of Gamma from 0 to T and plot the resulting optimal cost, you get the price of robustness curve.

1

Flat at low Gamma

Going from Gamma = 0 to Gamma = 2 barely increases cost. You get significant protection very cheaply. This is the region worth being in.

2

Steep at high Gamma

Going from Gamma = 8 to Gamma = T increases cost dramatically. The last few units of protection are very expensive.

3

The elbow point

Most curves have an elbow where flat transitions to steep. This is the natural Gamma to choose — most of the protection at a fraction of the maximum cost.

Deliverable

Run the Gamma sweep

Run the code and print the cost at each Gamma value from 0 to T.

Find the elbow

Look at the price of robustness curve. Find the elbow point. What Gamma value would you recommend? How much extra cost does it add compared to Gamma = 0?